

Site Area (ODA) 11,412,825.4 ■	Green Ratio 27.40%	Public Space Ratio 4.40%	Scenarios (Validation) 12 (3)	Handover Stations 9 Nodes
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1. Philosophy & Three Laws

- [Confluence Concept] Robots replace humans in hazardous aloft, confined, and toxic inspections; humans retain authorization, halt, and acceptance sovereignty.
- [Three Laws of Confluence] 1. Physical Labor Substitution; 2. Human Sovereign Final Review; 3. Public Space Pedestrian Priority.
- [Planning Paradigm] '50-Year Long-Life Spatial Frame + Fast Modular AI Infill', resolving lifecycle mismatch between 6-month AI shifts and 50-year structures.

2. Spatial Framework & Kiosks

- [One Spine] 8.7km Centennial Heritage Park Confluence Vitality Spine running north-south;
- [Three Fields] Zhongzhiyuan Arena (192.1ha) + AI Origin Garden (104.3ha) + Dazhongsi Market (72.0ha);
- [Twin Loops] 8.7km Human Greenway Loop + 7.2km Robot Low-Speed Service Twin Loops;
- [Nine Stations] 9 standardized Handover Kiosks (Type A Hub 1200■, Type B 300-500■, Type C 80-120■);
- [Four Landmarks] Centennial Switch Plaza, Handover Hub, Watchtower Drone Nest, Green-Wave Crossway.

3. Regional Loop & Governance

- [Regional Loop] Linking Future Science City (Energy AI), Huairou (Compute), and E-Town (Embodied Robotics).
- [Conflict Resolution] 1.5m crosswalk yield island, 1.2m courier delivery narrow lanes, 60cm blind-path buffer.
- [Governance] Certified On-Duty Voucher System & 22:00-06:00 off-peak night maintenance timetable.

4. Spatial Network Topology & Handover Kiosks Map

Full Confluence Spine & Handover Kiosks · 8.7km Human Greenway + 7.2km Robot Service Loops

Legend: — Heritage Spine (Gold) ··· Human Greenway (Orange) — Robot Loop (Teal) ● Handover Kiosks (S1-S9) ★ Landmarks

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1. Land Use Structure	2. Organic Renewal Strategy	3. Control Metrics Integrity
● Transport Road (1207): 365.2 ha (32.0%) - Arterial grid and twin loops ● Commercial/Service (09): 258.9 ha (22.7%) - Modern business & AI retail ● Reserved/Study (1301): 202.4 ha (17.7%) - Flexible innovation reserves ● Research & AI (08): 182.6 ha (16.0%) - Full-stack independent R&D ● Culture (0803): 77.6 ha (6.8%) - Wudaokou heritage interface ● Residential (07): 42.2 ha (3.7%) - Livable talent neighborhoods ● Green Space (14): 12.4 ha (1.1%) - Core parks & pocket green spaces	● [Retain] Preserve historic sites (Qinghuayuan Station) and quality residential communities; ● [Renovate] Upgrade inefficient industrial buildings in Zhongzhiyuan & commercial facades in Dazhongs; ● [Demolish] Strictly limited to dilapidated temporary structures to open walking corridors; ● [Construct] Build high-standard AI R&D spaces on verified potential development land.	● [Known Metrics] ODA Area 11.41 km², Green Ratio 27.40%, Public Space Ratio 4.40%, Building Footprint 3.71M ■ (54 blocks); ● [Unknown Disclosures] Floor area ratio, building height, and density kept unknown/null pending official controls; ● [Trigger] Full automatic package recalculation scheduled upon official regulatory data release.

4. Spatial Network Topology & Handover Kiosks Map
Full Confluence Spine & Handover Kiosks - 8.7km Human Greenway + 7.2km Robot Service Loops
S1 Qinghe N Waterfront Garden S2 Zhongzhiyuan Watchtower Hub S3 Houchangcun E Battery Swap Post S4 Tsinghua E AI Incubation S5 Wudaokou Handover Hub S6 Xueyuan Rd Reskilling Center S7 Zhichunlu Edge Isolation Pod S8 Dazhongs Green Wave Crossway S9 Jimenqiao E Logistics Micro-Post Legend: — Heritage Spine (Gold) - - Human Greenway (Orange) — Robot Loop (Teal) ● Handover Kiosks (S1-S9) ★ Landmarks

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1. Road Network & Twin Loops

- [Road Network] 50.6 km arterial network enhancing east-west stitching and microcirculation;
- [Slow Mobility Net] 16.9 km comprehensive walking-and-cycling network connecting transit and campuses;
- [Human Greenway Loop] 8.7 km continuous scenic trail along the heritage park and river corridors;
- [Robot Service Loops] 7.2 km north and south loops dedicated to delivery, inspection, and cleaning.

2. AI Traffic & Conflict Prototypes

- [Green-Wave AI Demo] Dazhongsi adaptive signal coordination with transit priority, police physical override & SWB fallback;
- [Crosswalk Yield] 1.5m robot deceleration island, radar 2m detection, slowing to 2 km/h static yield;
- [Commercial Narrow Lanes] 1.2m dedicated delivery lanes with 15cm sloped curb separation from courier scooters;
- [Blind-Path Buffer] 60cm continuous barrier-free buffer along tactile paving, guide robots 50cm lateral offset.

3. TOD Integration & Edge Infrastructure

- [TOD Integration] Underground-to-ground seamless transit integration at Wudaokou, Dazhongsi, and Xizhimen;
- [New Utilities] Handover stations integrate rooftop solar, battery swap kiosks, edge compute, and utility monitoring;
- [Low-Altitude Corridors] Watchtower drone hub and AED emergency routes under sandboxed airspace.

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S1 Qinghe N

S2 Zhongzhiyuan

S3 Houchangcun E

S4 Tsinghua E

S5 Wudaokou

S6 Xueyuan Rd

S7 Zhichunlu

S8 Dazhongsi

S9 Jimenqiao E

Waterfront Garden

Watchtower Hub

Battery Swap Post

AI Incubation

Handover Hub

Reskilling Center

Edge Isolation Pod

Green Wave Crossway

Logistics Micro-Post

Legend: — Heritage Spine (Gold) - - Human Greenway (Orange) — Robot Loop (Teal) ● Handover Kiosks (S1-S9) ★ Landmarks

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1. Blue-Green System & Metrics

- [Green Space Area] 3,127,039.4 ■ (312.7 ha), Green Ratio 27.40% (recomputed under EPSG:4548);
- [Public Space Area] 502,225.7 ■ (50.2 ha), Public Space Ratio 4.40% (core plazas and landmarks);
- [Water Network] Continuous blue-green corridors along Qing River (north) and Xiaoyue River (east);
- [Ecological Spine] Heritage Park spine stitches historical railway rifts into a unified urban fabric.

2. Four AI Pilgrimage Landmarks

- [Centennial Switch Plaza] Historic switchman sculpture side-by-side with modern inspection robots;
- [Human-Robot Handover Hub] Glass-domed visual authorization hall demonstrating transparent AI governance;
- [Watchtower] Composite drone nest and public viewing deck serving both inspection hub and citizen park;
- [Green-Wave Crossway] Four-quadrant walking linkage with interactive data-flow pavement lighting.

3. Components & Humanistic Care

- [Component Library] Modulated plazas, pocket parks, waterfront boardwalks, and station forecourts;
- [Barrier-Free Care] Strict compliance with Barrier-Free Law Art. 39, retaining physical human assistance;
- [Urban Character] Keynote of 'Heritage Humility × Future Feel × Transition Coordination'.

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1. Zhongzhiyuan Arena (192.1 ha)	2. Origin Community Garden (104.3 ha)	3. Dazhongsi Smart Market (72.0 ha)
[Role] Inspection robot testbed and national AI autonomous innovation cluster; [Structure] R&D blocks + hazardous operation live testbeds + Watchtower drone hub; [Projects] Power-line robot climbing field, utility tunnel inspection pod, Qinghe riverside garden; [Mobility] 5th-ring pedestrian bridge, segregated robot testing tracks and citizen paths; [Risk] Mandatory safety and environmental assessments for live robot testing grounds.	[Role] Fundamental AI research incubation and worker career-transition academy; [Structure] University tech incubation + Handover Hub + Switch Plaza + Healing Garden; [Projects] Railway heritage window, co-pilot reskilling workshop, hacker commons; [Mobility] Wudaokou & Tsinghua East Station transit integration, continuous greenway trail; [Risk] Strict preservation buffer for Qinghuayuan Station historic building.	[Role] AI-native retail consumption and modern commercial services cluster; [Structure] AI commerce blocks + dedicated delivery narrow lanes + Green-Wave Crossway; [Projects] Smart device experiential market, four-quadrant transit linkage, interactive light plaza; [Mobility] 1.2m delivery robot lanes segregated from pedestrian paths, organized courier parking; [Risk] Visual corridor preservation for historic Dazhongsi Temple.

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1. 12 Scenario Cards Matrix

- 1. Power-Line Partner (Zhongzhiyuan/Validation): thermal map / edge anonymized / 7d / SWB manual power check
- 2. Utility Culvert Drone (Culvert/Validation): toxic gas & thickness / enclosed / 30d / SWB ventilation entry
- 3. Traffic Green-Wave AI (Dazhongsi/Validation): queue flow / face blurred / 24h / police override / SWB fixed
- 4. Bridge Health Robot (Zhongzhiyuan): strain & cracks / no privacy / 30d / licensed engineer / SWB manual
- 5. Fire Recon Drone (Airspace): thermal matrix / encrypted / incident log / commander / SWB scout team
- 6. Night Cleaning Robot (Loops): LiDAR / 22:00-06:00 off-peak / 24h / sanitation platform / SWB daytime
- 7. Guide-Cane Robot (Greenway): ultrasonic ranging / zero biometrics / memory clear / SWB white cane & volunteer
- 8. Delivery Robot (Dazhongsi): narrow lane / face blurred / 2h purge / merchant / SWB courier direct
- 9. Emergency AED Drone (Corridor): GPS coordinates / zero medical privacy / 7d / 120 ER / SWB ambulance
- 10. Construction Safety AI (Sites): hard-hat detection / no face DB / 14d / supervisor / SWB manual inspection
- 11. Transit Feeder Shuttle (Station): passenger volume / edge blurred / 7d / safety steward / SWB standard bus
- 12. Senior Greeting Robot (Community): voice keyword / zero video / 24h / grid officer / SWB social worker visit

2. Data Privacy & Sovereignty

- [Edge Anonymization] Compliance with Personal Information Protection Law; real-time Gaussian face blurring;
- [Service Withdrawal Baseline (SWB)] 100% manual operational fallback and hardware physical E-stop switches;
- [Complaint Channel] 15-minute community grid & 12345 joint response for public inquiries and complaints.

3. Personas & Operations

- [6 User Personas] Electrician Zhang, Traffic Assistant Li, Engineer Wang, Docent Chen, Resident Liu, Courier Zhou;
- [Flagship Events] Annual Human-Robot Confluence Congress + Developer Simulation Challenge + Scenario Access;
- [International Story] 'From Switchman to Co-Pilot', conveying human-centered technological modernization.

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1. Core Metrics Recalculation

- Site Area (ODA): 11,412,825.4 ■ (11.41 km² · +0.11% vs brief) · known (provisional)
- Green Area / Ratio: 3,127,039.4 ■ (27.40%) · strict EPSG:4548 projection
- Public Space Area / Ratio: 502,225.7 ■ (4.40%) · 6 core nodes
- Building Footprint Area: 3,710,756.5 ■ (54 conceptual blocks)
- Road Network Length: 50,560.8 m (50.6 km)
- Slow Mobility Network: 16,864.2 m (16.9 km)
- FAR / Height / Density: unknown / null (honest boundary disclosure pending controls)

2. Four Self-Check Quality Gates

- [Deterministic Validation] 43 files declared in manifest, SHA-256 chain 100% matched;
- [Spatial Review] EPSG:4548 projection, 99.9998% zero-overlap topological coverage;
- [Visual Packaging] 14 keyword anchors matched, 3 core metrics data-values aligned;
- [Professional Evidence] 9 standards (5 mandatory addressed), 15 design depth items closed;
- [Preflight Check] Package size 3.2 MiB (far below 50 MiB limit), 0 blockers 0 warnings.

3. Font Licensing & Disclaimers

- [Font Licensing] Web uses client sans-serif, PDFs use standard CID sets, figures rasterized with open fonts;
- [Status] Conceptual proposal for open competition; does not constitute statutory approval or fiscal commitment;
- [Recomputation] Full recomputation scheduled upon release of official statutory redline and control files.

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AI Incubation

S5 Wudaokou

Handover Hub

S6 Xueyuan Rd

Reskilling Center

S7 Zhichunlu

Edge Isolation Pod

S8 Dazhongsi

Green Wave Crossway

S9 Jimenqiao E

Logistics Micro-Post

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